the hole with a jet and an accretion disk. Gas from the disk falls into and through the horizon, and that infalling gas dominates what the camera sees at and beneath the horizon. In *Interstellar*, by contrast, there is no jet, and the accretion disk is so anemic that it is not currently sending any of its gas into and through the horizon, so the hole's interior looks rather dark. However, in *Interstellar* Cooper encounters a dim fog of light and white flakes from stuff that fell in before him. These are not the result of simulations, but instead were put in by hand by the Double Negative artists.

Chapter 29. The Tesseract

When Christopher Nolan told me he was going to use a tesseract in *Interstellar*, I was delighted. At age thirteen I read about tesseracts in Chapter 4 of George Gamow's marvelous book *One, Two, Three, . . . Infinity* (Gamow 1947), and that had a major role in making me want to become a theoretical physicist. You can find a detailed discussion of tesseracts in *The Visual Guide to Extra Dimensions* (McMullen 2008). Christopher Nolan's complexified tesseract is unique; there is not yet any public discussion of it anywhere, except in this book and others connected to the movie *Interstellar*.

In Madeleine L'Engle's classic science fantasy novel for children, *A Wrinkle in Time* (L'Engle 1962), children travel via a tesseract—they "tesser"—to find their father. My own interpretation of this is a journey through the bulk, riding in the face of a tesseract, like my interpretation of Cooper's trip from Gargantua's core to Murph's bedroom, Figure 29.4.

Chapter 30. Messaging the Past